exchanging information between one event and another. Reality is a network of granular events; the dynamic which connects them is probabilistic; between one event and another, space, time, matter and energy melt in a cloud of probability.

This strange new world is slowly emerging today from the study of the main open question posed in fundamental physics: quantum gravity. It's the problem of coherently synthesizing what we have learned about the world with the two major discoveries of twentieth-century physics: general relativity and quantum theory. To quantum gravity, and the strange world that this research is unfolding, this book is dedicated.

The book is a live coverage of the ongoing research: what we are learning, what we already know, and what we think we are beginning to understand, about the elementary nature of things. It starts from the distant origin of some key ideas that we use today to order our understanding of the world and describes the two great discoveries of the twentieth century – Einstein's general relativity and quantum mechanics – trying to put into focus the core of their physical content. It tells of the picture of the world which is emerging today from research in quantum gravity, taking into account the latest indications given by nature, such as the confirmation of the cosmological standard model obtained from the Planck satellite and the failure at CERN to observe the super-symmetric particles that many expected. And it discusses the consequences of these ideas: the granular structure of space; the disappearance of time at small scale; the physics of the Big Bang; the origin of black-hole heat – up to the role of information in the foundation of physics.

In a famous myth related by Plato in the seventh book of *The Republic*, some men are chained at the bottom of a dark cave and see only shadows cast upon a wall by a fire behind them. They think that this is reality. One of them frees himself, leaves the cave and discovers the light of the Sun, and the wider world. At first the light, to which his eyes are unaccustomed, stuns and confuses him. But eventually he can see, and returns excitedly to his companions to tell them what he has seen. They find it hard to believe.

We are all in the depths of a cave, chained by our ignorance, by our prejudices, and our weak senses reveal to us only shadows. If we try to see further, we are confused: we are unaccustomed. But we try. This is science. Scientific thinking explores and redraws the world, gradually offering us better and better images of it, teaching us to think in ever more effective ways. Science is a continual exploration of ways of thinking. Its strength is its visionary capacity to demolish preconceived ideas, to reveal new regions of reality, and to construct novel and more effective images of the world. This adventure rests upon the entirety of past knowledge, but at its heart is change. The world is boundless and iridescent; we want to go and see it. We are immersed in its mystery and in its beauty, and over the horizon there is